

SigFJProp: Lightweight and Scalable Signed Graph Learning via Opinion Dynamics

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Background

Signed Graph Learning

- Signed graph aims to model networks where edges encode both positive and negative relationships, such as trust and distrust, agreement and disagreement.
- Signed graph learning aims to learn effective node representations by modeling the positive and negative relationships between nodes, thereby enabling [link sign prediction](#).

Motivation

- Most signed graph neural networks (SGNNs) rely on deep architectures with multi-hop message passing and complex propagation schemes. These designs incur significant computational overhead on large-scale networks.
- The classical Friedkin-Johnsen (FJ) model provides important inspiration for our work, as opinion evolution in social networks closely resembles the message-passing mechanism in graph neural networks.

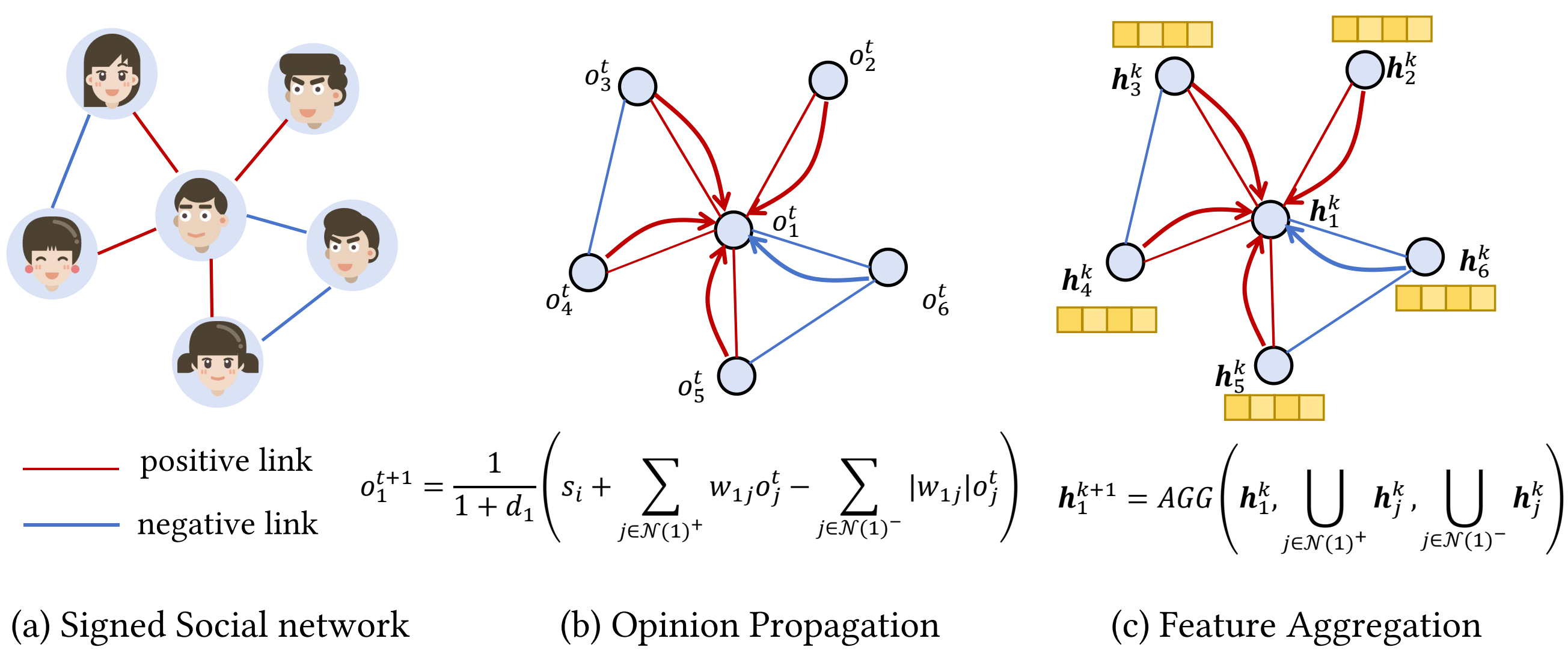


Figure 1. (a) A signed social network. (b) Opinion propagation process in Friedkin-Johnsen model. (c) Feature aggregation in GNN.

Contribution

- We establish a formal connection between signed message-passing in GNNs and opinion dynamics, and introduce [SigFJProp](#), a principled and [lightweight](#) SGNN model grounded in the Friedkin-Johnsen process.
- We develop a scalable linear-time approximation method [EFS](#) for computing the signed forest matrix, and significantly accelerate the process via [parallelization](#), enabling efficient inference on large-scale signed networks.
- We conduct comprehensive experiments on multiple benchmark datasets, including the largest signed graph evaluated to date—with [over 750K nodes and 10 million edges](#)—demonstrating that SigFJProp achieves competitive predictive performance while significantly reducing computational cost.

Proposed Model

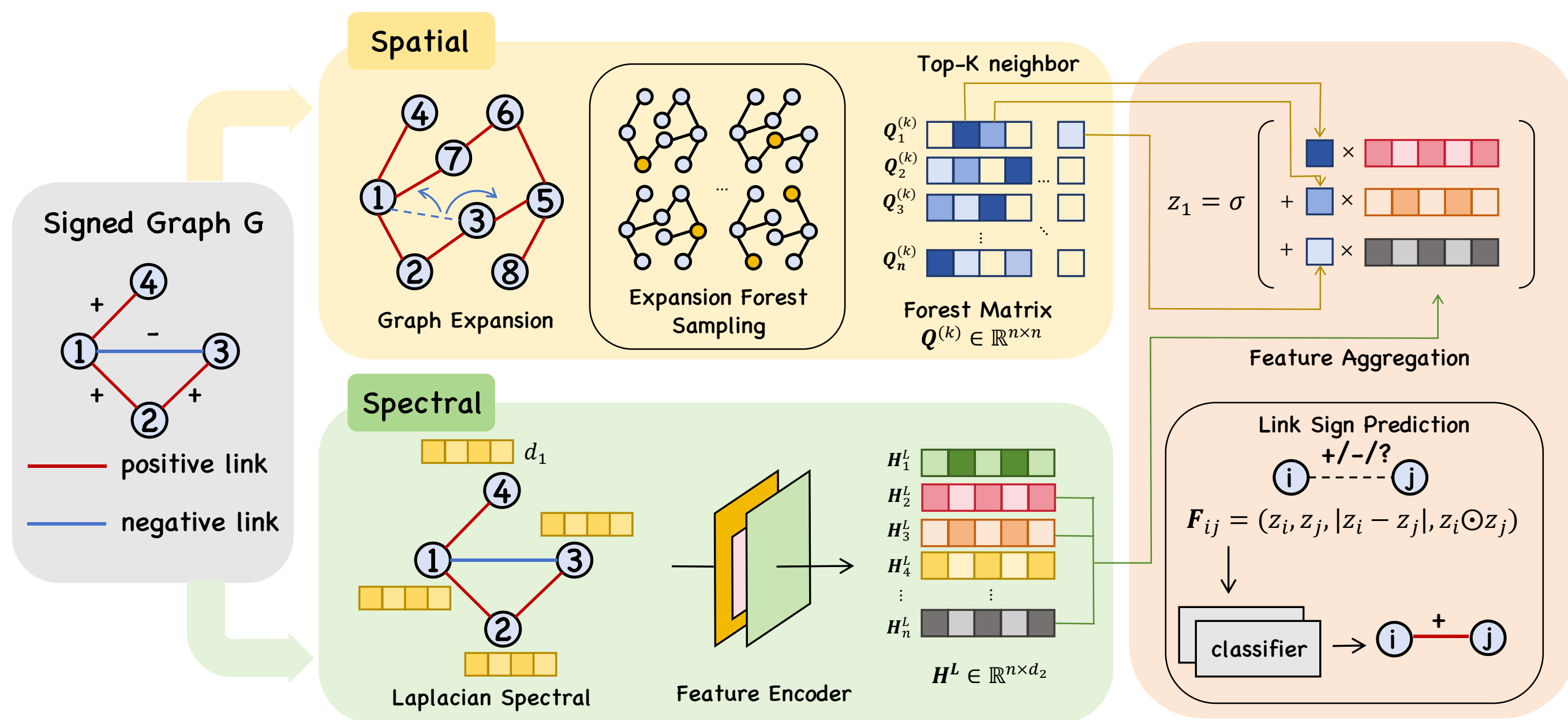


Figure 2. The SigFJProp Architecture.

Spectral Encoding

To obtain expressive node features, we first compute spectral embeddings by decomposing the signed adjacency matrix. Then, we introduce a Spectral Feature Encoder, denoted as a parametric function f_θ , which maps the input feature matrix $X \in \mathbb{R}^{n \times d_1}$ into a latent representation $H \in \mathbb{R}^{n \times d_2}$.

Spatial Encoding

We apply a spatial propagation mechanism inspired by the signed FJ model. Specifically, we introduce a residual connection to retain each node's initial features and propagate information through an influence-aware aggregation. The update rule is defined as:

$$z_i = \sigma \left(\sum_{j \in V} (I + D - W^+ - W^-)^{-1}_{ij} H_j \right).$$

- To reduce computational cost, we find that for each node i , it is sufficient to select only the [top-K nodes](#) in V for propagation.
- We define the matrix $Q = (I + D - W^+ - W^-)^{-1} = (I + D - W)^{-1}$, and approximate it by truncating each row of Q , keeping only the top-K entries in absolute value.

Approximation Algorithm for Acceleration

Graph Expansion

- For a signed graph $\mathcal{G} = (V, E = E^+ \cup E^-, \sigma)$, we construct a new unsigned graph $\tilde{\mathcal{G}} = (\tilde{V}, \tilde{E})$ with $2n$ nodes and $2m$ edges. For the corresponding unsigned graph $\tilde{\mathcal{G}}$, we label its $2n$ nodes as $1, 2, \dots, 2n$. For each edge $(i, j) \in E$, if $\sigma_{(i,j)} = +1$, we add two edges (i, j) and $(i + n, j + n)$ to $\tilde{E}$. And if $\sigma_{(i,j)} = -1$, we add $(i, j + n)$ and $(i + n, j)$ to $\tilde{E}$. $\tilde{\mathcal{G}}$ is the *expansion graph* of $\mathcal{G}$.
- For a signed graph $\mathcal{G}$ and its corresponding expansion graph $\tilde{\mathcal{G}}$:

$$Q_{n \times n} = \frac{1}{2} [I \ -I] \tilde{Q}_{2n \times 2n} \begin{bmatrix} I \\ -I \end{bmatrix}.$$

Expansion Spanning Forest Interpretation

Let $\mathcal{F}$ denote the set of spanning forests for $\tilde{\mathcal{G}}$ and $\mathcal{F}_{ij}$ denote the set of spanning forests where node i is rooted at j . Then, for any $i, j \in V$, we have

$$Q_{ij} = \frac{|\mathcal{F}_{ij}| - |\mathcal{F}_{i,j+n}|}{|\mathcal{F}|}.$$

- Algorithm EFS samples l forests in $\tilde{\mathcal{G}}$, and runs in $O(m + nl)$ time and provides an (ϵ, δ) -approximation for the entries of Q .
- The spanning forests generated in EFS algorithm are mutually independent, the sampling process is inherently amenable to [parallelization](#) (EFS-P).

Experiment

AUC	Method	BitcoinAlpha	BitcoinOTC	WikiElec	WikiRfa	Slashdot	Epinions	TwitterSG
Statistics	#Nodes	3,783	5,881	7,115	11,017	82,140	131,828	753,944
	#Edges	24,186	35,592	103,689	170,335	549,202	841,372	12,848,093
Unsigned GRL Methods	GCN	0.665±0.003	0.742±0.004	0.713±0.006	0.721±0.004	0.685±0.011	0.784±0.013	0.683±0.005
	GAT	0.681±0.005	0.766±0.005	0.735±0.006	0.748±0.004	0.703±0.010	0.801±0.009	-
	GIN	0.659±0.003	0.738±0.004	0.710±0.005	0.716±0.003	0.679±0.009	0.780±0.009	-
	PPRGo	0.675±0.002	0.759±0.003	0.728±0.003	0.737±0.003	0.698±0.005	0.795±0.007	0.665±0.004
	SGFormer	0.724±0.002	0.802±0.002	0.762±0.003	0.774±0.002	0.742±0.004	0.836±0.006	0.719±0.003
Signed GRL Methods	SGCN	0.708±0.001	0.780±0.002	0.811±0.008	0.825±0.004	0.702±0.016	0.880±0.017	0.763±0.005
	STNE	0.844±0.001	0.879±0.003	0.820±0.016	0.818±0.012	0.835±0.004	0.901±0.008	-
	SNEA	0.872±0.004	0.895±0.005	0.816±0.003	0.813±0.004	0.842±0.003	-	-
	SGCL	0.855±0.001	0.917±0.004	0.827±0.005	0.837±0.003	-	-	-
	SLGNN	0.792±0.003	0.879±0.004	0.792±0.004	0.833±0.003	0.806±0.002	-	-
Ours	SE-SGFormer	0.870±0.002	0.794±0.002	0.837±0.002	0.850±0.002	-	-	-
	SigFJProp	0.912±0.002	0.916±0.002	0.843±0.002	0.856±0.002	0.857±0.003	0.915±0.003	0.806±0.001

Figure 3. Model Performance Comparison

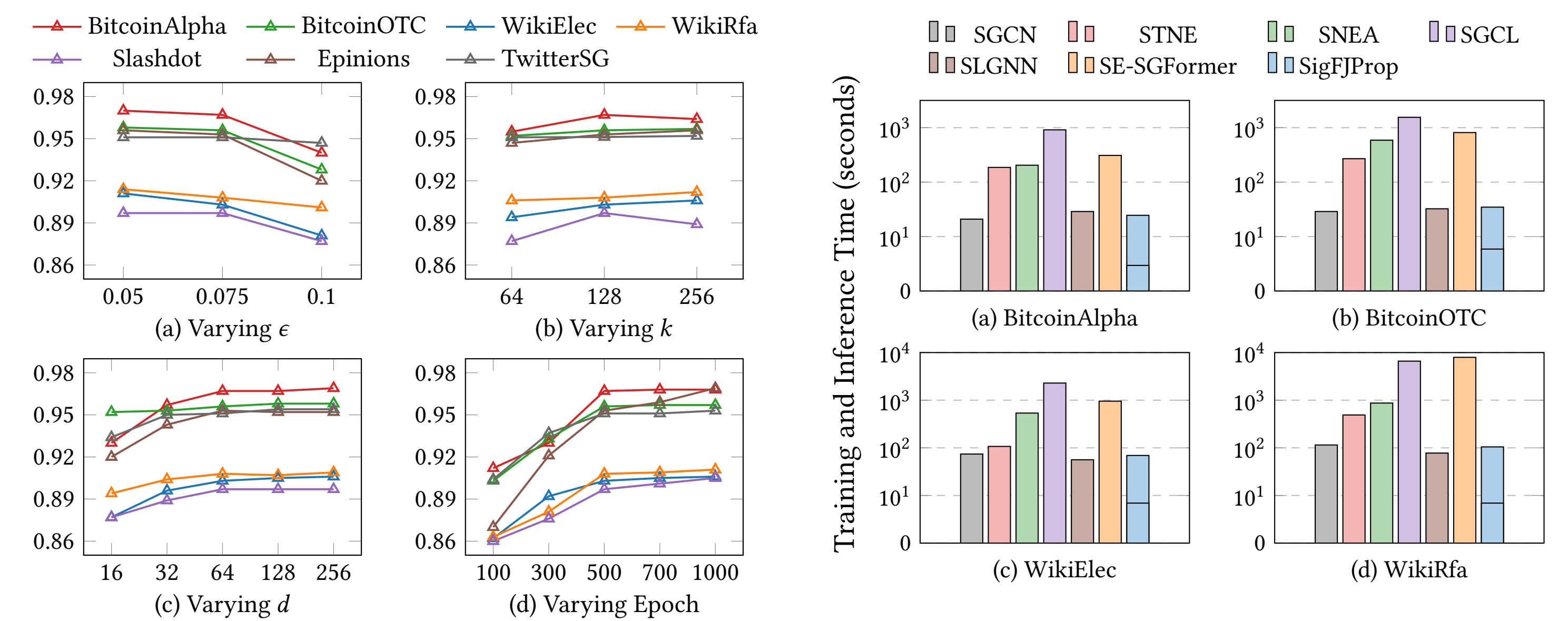


Figure 4. Hyperparameter sensitivity analysis.

Figure 5. Training and inference time of signed GRL models.

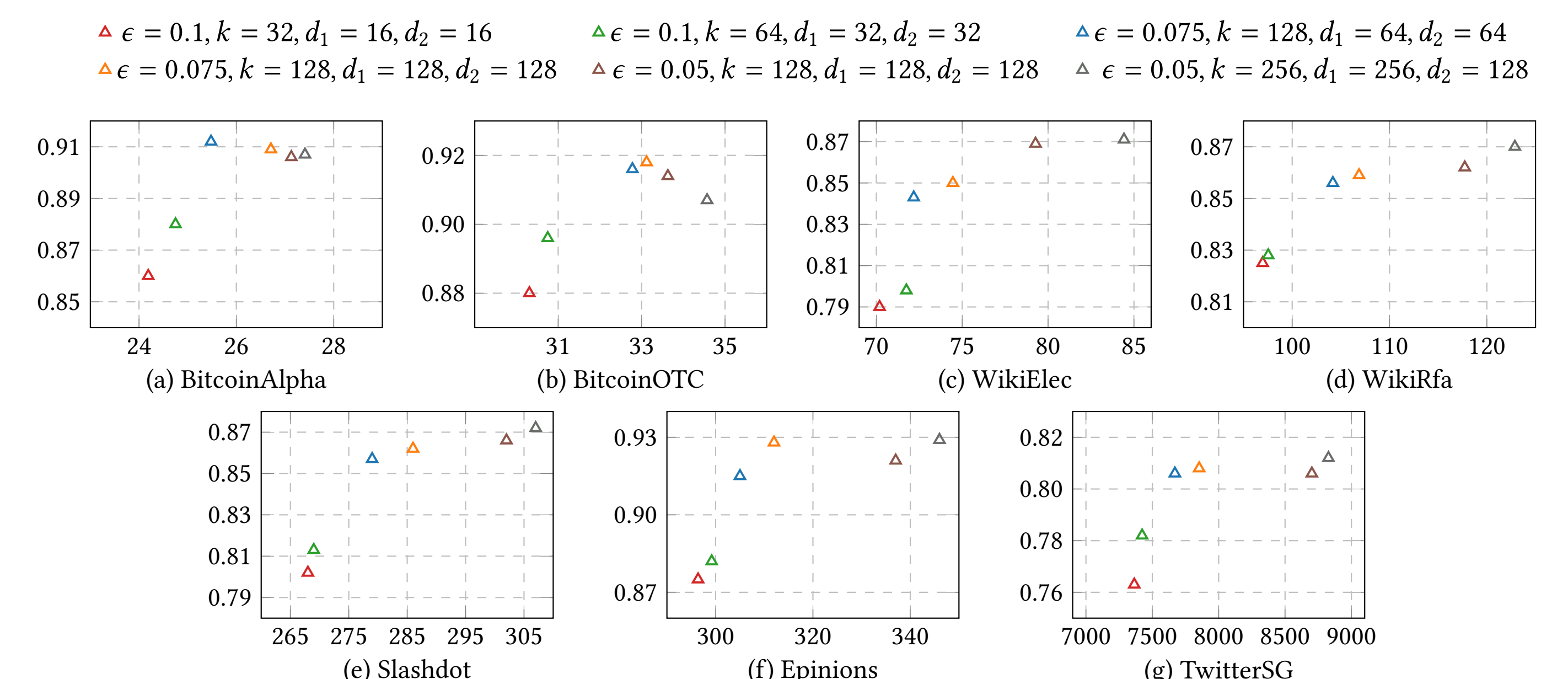


Figure 6. Trade-off analysis.